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BUSINESS AND PROJECT MANAGEMENT

AlmReady

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1. Problem/Solution Fit

Many students underperform in oral examinations due to performance anxiety and the absence of objective feedback. To address this challenge, AlimReady introduces an AI-powered virtual examination panel that realistically simulates the pressure and dynamics of a real oral exam. By combining voice, video, and multi-agent AI analysis, the system extracts objective behavioral metrics, evaluating speech quality, pacing, confidence, and non-verbal behavior to help students improve their performance.

1.1 Information Sources and Methods Used

1.1.1 Academic Search

To build this work on solid ground, we ran a structured search on Scopus. Open community sources are covered in Section 1.1.3.

To keep our searches focused and avoid noise, we followed an iterative process on the academic side:

- Definition of keywords related to our core topics: public-speaking anxiety, multimodal analysis, agent-based models and AI/VR tutoring.
- Construction of an initial query using the operators AND (to combine required concepts) and OR (to add synonyms).
- Analysis of the retrieved results to spot irrelevant matches (noise), added back into the query with the AND NOT operator.

For each query the table shows the search string, how many results it returned, and why we ran it: what we wanted to verify and what information we were looking for.

Step	Search string (TITLE-ABS-KEY on Scopus)	Results	Why we ran it
Q1 The problem is real	("public speaking anxiety" OR "glossophobia" OR "exam anxiety" OR "oral exam stress") AND ("student performance" OR "academic failure" OR "evaluative stress")	22	To confirm that exam and speaking anxiety is a real problem that hurts student performance, and to see how teachers already deal with it.
Q2 Virtual agents / avatars for speaking practice	("virtual agent*" OR "virtual human*" OR "agent-based model*" OR avatar) AND ("public speaking" OR "interview training" OR "oral presentation" OR "communication skills")	338	To check whether virtual agents and avatars are already used to train speaking skills.
Q3 Multimodal analysis of oral presentations	"multimodal analysis" AND "oral presentation"	6	To verify that combining speech and behavioral signals to give feedback on oral performance is technically feasible and already tested.

Q4 Public-speaking anxiety in students	"public speaking anxiety" AND students	191	To gather evidence on how wide anxiety is spread is among students and whether AI or VR practice is being tested to reduce it.
Q5 Edge AI + biometrics under privacy constraints	("edge computing" OR "edge AI") AND ("facial expression" OR "speech recognition" OR "biometric*") AND (privacy OR "data protection" OR gdpr) AND NOT "IoT"	152	To learn how to handle sensitive signals like facial or voice data directly on the device, to stay within European regulation and keep our privacy-friendly design.
Q6 AI / VR tutors for oral-exam preparation	("virtual reality" OR "VR" OR "AI tutor*" OR "simulator") AND ("public speaking training" OR "oral exam preparation") AND NOT "medical education"	51	To find whether AI or VR tutors built specifically for oral-exam preparation already exist and how effective this kind of tools are.

These are only some examples of the query we used that, together with other research methods, allowed us to obtain all of the necessary information for our study.

1.1.2 Patent Search and Freedom to Operate

The academic search above establishes that the problem is real and the technology is feasible. It does not, however, tell us whether someone already owns the right to build what we are proposing.

We searched Lens.org and Espacenet, moving iteratively from a broad classification to a narrow, inspectable set. The classification used is CPC/IPC G09B 19/04, which is where teaching systems for spoken skills are filed.

Step	Query	Results	What it tells us
P1	G09B19/04	3,688	The field of speech-teaching apparatus is large and mature
P2	P1 + ("artificial intelligence" OR "LLM" OR "machine learning")	244	AI-based speech teaching is an active, populated area
P3	P2 + ("anxiety" OR "stress") AND ("oral exam" OR "interview")	14	Our specific intersection is small enough to inspect by hand

The drop from 3,688 to 244 shows that AI has entered this field but does not dominate it. The drop from 244 to 14 shows that our specific combination (AI, anxiety, and an examination context) is a narrow argument inside a crowded field.

We read the 14 results and identified three US patents: US 5870709 A (Bernstein, 1996, expired) [1], US 11798431 B2 (Pitchvantage, 2014, active in the US) [2], and US 10818193 B1 (Aptima Inc, 2018, active in the US) [3]. We also found one further EU application that matters more than all three: EP 4075411 A1 (Samsung Electronics, priority 2020, pending, examination in progress) [4].

Nothing blocks us. Patents are territorial, and the family jurisdictions of all three US documents are US-only. They have no effect in our market.

Even setting jurisdictions aside, our system falls completely outside their claims.

Samsung's EP application is the one document we monitor, but it claims a virtual audience asking questions built from entities extracted through relationship templates; we provide an examination panel, and our agents reason over the uploaded material rather than filling templates. Moreover, it is still under examination.

No patent covers our use case. Nothing addresses a subject-aware examination for oral exams with no scores and no emotion inference.

1.1.3 Open Community Sources

The academic evidence in Section 1.1.1 establishes that oral-exam anxiety is a documented phenomenon. It does not, however, tell us whether Italian students themselves perceive it as a problem worth solving. To capture direct user perceptions, we complemented the academic search with a screening of open community sources, focusing on r/Universitaly, the main Italian-language student community on Reddit, where students discuss university life spontaneously and without any research framing.

We used the query "ansia esame orale". The search returned approximately 160 posts, which were screened using a combination of manual review and Generative AI-assisted content classification. We identified 33 posts explicitly related to oral-exam anxiety or preparation difficulties, while 15 additional posts discussed exam anxiety in more general terms. The relevance of this topic is illustrated by highly engaged discussions such as:

1	<i>"Consiglio: come faccio ad essere più tranquilla per gli esami orali? Letteralmente le cose le so solo che mi ispanico [impanico] e mando tutto all'aria"</i>
2	<i>"Gli esami orali mi distruggono"</i>
3	<i>"Odio gli esami orali"</i>
4	<i>"Ma ha senso fare esami orali?"</i>
5	<i>"Primo esame universitario e l'ansia mi sta mangiando viva"</i>

This search was conducted to verify whether oral-exam anxiety emerges as a spontaneously discussed pain point among Italian university students, outside any research or commercial context.

The Reddit discussions provide qualitative evidence consistent with the findings from the academic literature. Threads describing oral-exam anxiety are frequent, commented, and recent. Two recurring patterns are especially relevant to our positioning. First, students explicitly report that they know the material but perform poorly because anxiety interferes with their delivery (number 1 of the list above), which is precisely the gap between preparation and delivery that AlmReady targets. Second, first-year students emerge as the segment where the distress is most acute (number 5 of the list above), providing additional support for the segmentation choice made in Section 4.2.1.

1.2 Problem Definition

Fear of public speaking is one of the most prevalent forms of social anxiety and is consistently reported as the most commonly feared situation in the general population. Research suggests that up to one-third of individuals experience significant public-speaking anxiety, with negative consequences for academic performance, professional development, and social interactions [5].

If we zoom in on university students specifically, a 2026 study on 210 first-year health-science students found that 74.8% reported moderate stress before exams, and another 14.3% reported high stress. Only 11% reported low stress [6]. So for this category of people, feeling stressed before an exam is basically the norm, not the exception.

Two more studies matter a lot here. First, a study with 42 students on a Speech course found that public-speaking anxiety really hurts both grades and future job chances [7]. The program used to help students already included Yoodli (see section 2.1.1), a generic AI coaching app, as one of its tools. This tells us teachers are already reaching for generic AI apps to fix this problem, but nobody has built a tool made specifically for oral exams yet. Second, a study called Corp-Oral tested 36 university students with repeated, structured practice sessions and feedback. It lowered self-reported anxiety by 33.2% [8]. So we know repeated structured practice genuinely works, even without any AI involved.

Right now, what students actually do to prepare is very basic: practice alone in front of a mirror, get quizzed by a parent or a friend, or just repeat the material in their head. None of this gives real feedback. There's no score, no way to measure hesitation or vocal stress, and no one acting like a real examiner who can interrupt or react.

1.3 AI-Enabled Solution

Our project, AlmReady, is a virtual exam panel built as a multi-agent system. Park et al. [9] showed that LLM-based agents can behave in believable, coherent ways over time (remembering past events, reflecting on them, and using that to plan what to do next) well beyond just answering single questions. We borrow that same three-part underlying architecture

(observation, reflection and planning), but repurpose it: instead of agents living an open-ended simulated life, our agents are examiner personas that stay focused on one task: interrogating the student on a specific subject. In our case, the examiner agent observes what the student has said so far, reflects on it to judge understanding and plans the next question based on that judgment (for example, asking a targeted follow-up when it notices a gap, instead of just moving down a fixed list of questions).

The three agents have a distinct role and a distinct voice. The first agent (built on GPT) acts as the lead examiner, asking the main subject-matter questions in a calm, structured way. The second agent (built on Llama) acts as a faster-paced assistant examiner, jumping in with more direct follow-up questions. The two take turns: the lead examiner drives the main flow and the assistant steps in at set points to push harder. Both agents work from the live transcript of what the student has said so far. A third agent (built on Claude) doesn't speak to the student at all. It works as a flow controller: it reads two kinds of observable data, the content of the answers (from the live transcript) and measurable delivery metrics (pause duration, speech rate, filler-word frequency), and coordinates the other two agents through predefined pacing rules. For example, if pauses grow beyond a set threshold or answers become incomplete, it tells the lead examiner to simplify the next question and the assistant to hold its follow-ups. The coordinator never labels or infers the student's emotional state: it only applies pacing rules to measurable communication metrics, the same way a human examiner adjusts the rhythm of an exam.

Each model was chosen to fit its role. GPT leads the exam because it handles complex, structured reasoning well and keeps a coherent line of questioning over a long conversation. Llama runs the assistant examiner because it is fast and cheap to run, which fits a role built on quick, frequent follow-ups. Claude coordinates in the background because it is strong at following detailed behavioral instructions and staying consistent, which is exactly what the silent supervisor role needs.

To make the questions relevant to what the student is actually studying, the student can upload the course material and the examiner agents draw their questions from that specific content, instead of asking generic questions about the subject.

Along with the conversation layer, we add a multimodal analysis layer. Radford et al. [10] built Whisper, a speech recognition model trained on a huge amount of real-world audio, so it can transcribe speech accurately even with background noise, accents, or hesitant speech, not just clean, scripted audio. We use that same kind of model to turn the student's speech into text and flag things like pauses, filler words, and pacing. On top of that, a lightweight browser-based computer-vision model reads non-verbal signals straight from the student's webcam: gaze direction, gesture stability, and articulation rate.

This combination has already been tested with the RAP system [11]. It is an early example of a tool that gives automatic feedback on oral presentations, using posture, gaze, volume, and filled pauses from cheap sensors. In its first evaluation, RAP's automated feedback closely matched what real human evaluators said. That's direct proof that multimodal behavioral analysis can get close enough to expert human judgment to actually be useful, not just theoretically nice.

Recent literature further confirms this approach. A quasi-experimental study evaluating an AI-integrated VR oral-training application (20 university students) demonstrated significant reductions in both public-speaking and interview anxiety following a five-session intervention [12]. Combined with the RAP system findings above, this shows two things: AI can measure this kind of stress and AI-guided practice can actually reduce it.

A key design choice is the separation between the live exam and the evaluation phase. During the exam, all three agents run live (two speaking in turn, one coordinating silently in the background) creating a more realistic, multi-examiner panel feel. Once the exam is completed, a separate multi-agent panel, composed by GPT-4o-mini and Llama 3.1 8B (See section 3.2.3), rereads the full transcript and behavioral data to generate the final feedback report. The report is strictly limited to observable, communicative-behavior metrics and never includes any assessment of the student's emotional state. Specifically, it covers:

- content accuracy and completeness of the answers with respect to the subject matter;
- clarity and structure of the explanations;
- speech delivery indicators such as pacing, filler words, and vocal steadiness;
- non-verbal communication signals such as gaze direction and gesture stability.

For each of these dimensions the report provides a concrete, actionable suggestion for improvement (for example, "reduce filler words when answering unexpected questions" or "expand the definition before giving examples"). The feedback is always framed as communicative performance on the exam task, never as an inference about how the student felt.

This two-step approach follows the same principle used in the NYU case study [13]: live AI examiners during the exam, a separate panel responsible for the evaluation afterward.

Another key design choice comes from the regulatory review in Section 6. The system is built on purpose as a training tool whose output never becomes an official grade and the multimodal signal is described to users as "communicative-behavior feedback" (pacing, filler words, gaze, vocal steadiness) never as an inferred emotional state. This keeps the product inside the exemption the European Commission explicitly carved out for role-play and training simulations, explained in Section 6.2.

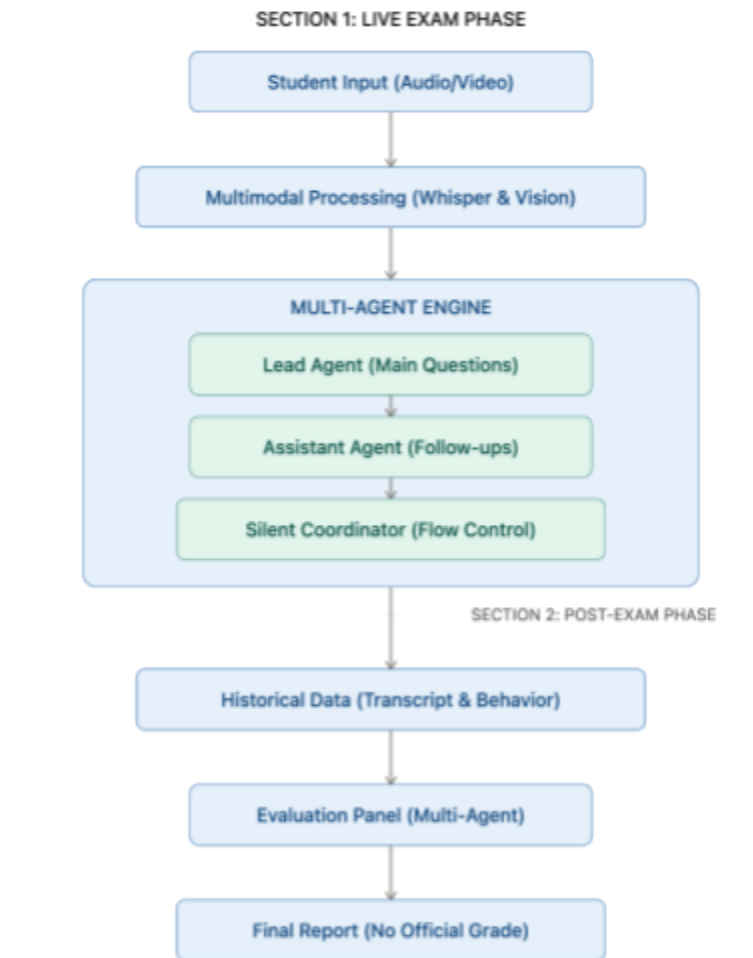


Figure 1. AlmReady system architecture: the live exam phase (multimodal capture and multi-agent engine) feeding into the post-exam phase (asynchronous evaluation and report generation).

1.4 Achieving Product-Market Fit

We're targeting two customer groups.

- The B2C group is individual students getting ready for an oral exam, a thesis defense, or a job interview.
- The B2B segment targets universities, offering them a measurable improvement in the outcomes: fewer failed oral exams and fewer students falling behind. It also helps them improve the overall quality of services they offer to students, which is something universities are increasingly evaluated on.

Compared to the alternatives analyzed in Section 1.2, our solution offers three things nobody else combines at once:

- a repeatable, objective set of behavior metrics
- real pressure from a simulated panel instead of a passive listener
- a cost structure that scales to numerous practice sessions, unlike a paid human tutor.

1.5 Logical Assumptions Foundation

- **Willingness to practice and pay:** students see exam anxiety as a solvable problem and are willing to pay for something that reduces it.
- **Real academic improvement:** lower exam anxiety and better practice actually lead to better exam performance: not just less stress, but real academic results, like fewer failed exams and better grades.
- **Technology transferability:** multimodal analysis can be moved to the oral-exam use case without needing a brand-new research breakthrough.
- **Believability of the simulated panel:** an LLM-driven examiner is realistic enough to trigger performance pressure, not just feel like chatting with a bot.
- **Proven idea, no real product yet:** the NYU project and some early papers show this works. Nobody has built an actual product for university oral exams.
- **Regulatory navigability:** the product stays legal in the EU only by staying strictly on the "training, non-certifying" side of the AI Act's emotion-recognition rules, explained in Section 6.2.

2. Competition Analysis

2.1 Competitive Analysis

2.1.1 Direct competitors

Yoodli[14] and Poised[15] are the closest commercial products: both use AI to analyze a webcam/mic feed during a speech or meeting and give feedback on filler words, pace, eye contact and tone. Neither one is built around an academic oral exam though, they coach generic public speaking or work meetings, they don't simulate an interrogating panel that asks follow-up questions on actual course content, and they don't try to recreate the pressure of a real exam.

2.1.2 The real prior-art case

The single most important finding in our whole competitive analysis is a documented, citable case: in late 2025, Panos Ipeirotis (NYU Stern) and Konstantinos Rizakos built an AI examiner called Viva [13], using ElevenLabs Conversational AI for the voice and a three-model "grading council". It was actually used on two real student cohorts (36 students Fall 2025, 37 Spring 2026), at a real marginal cost of about \$0.30-1 per exam. This proves that the core idea (a voice AI examiner plus a multi-LLM grading panel) already works, has already been tested twice and is already published. Our differentiation comes from its specialization for oral-exam preparation, behavioral feedback, and anxiety-reduction training.

2.1.3 Indirect competitors and substitutes

We found a cluster of VR-based public-speaking trainers:

- an immersive VR-based intervention designed to reduce public-speaking anxiety through gradual exposure to realistic speaking situations, currently being evaluated in a randomized controlled trial [16];
- an AI+VR oral-training app that, with 20 students, significantly cut both public-speaking and interview anxiety after five sessions [12] (already described in section 1.2);
- an LLM-driven VR simulation, tested specifically to measure how students accept and use the technology (UTAUT) [17];
- PAVE, a virtual reality platform that dynamically adapts audience reactions to help STEM students develop public-speaking confidence [18];
- and an LLM-driven chatbot training tool aimed for graduating students to mitigate Public-Speaking Anxiety (PSA) exploiting Question and Answer sessions [19].

All of these solutions pursue a similar objective: reducing public-speaking anxiety through repeated practice and exposure. However, they rely on dedicated VR hardware, making adoption more expensive and less accessible than a browser-based solution that only requires a standard webcam and microphone.

Beyond AI tools, the classic non-AI substitutes still matter: live public-speaking courses, private tutoring, faculty-run mock orals, and, for the worst cases, anxiety medication or therapy.

2.2 Assessing Competitors and Strategic Positioning

Yoodli and Poised are strong on brand and funding, but generic since they weren't built around the specific interrogation and anxiety dynamics of an oral exam. That leaves room for us, but it also means they could plausibly build an "exam mode" faster than us if they wanted to, since they already have the users and the infrastructure.

The NYU case is our real weak spot, not a strength: it shows the core mechanism (a voice AI examiner plus a multi-LLM panel) is already public and already published, so we can't claim it as an original idea.

Our strength has to be something else: the proprietary interaction dataset we build over time and the fact that we focus purely on anxiety-reduction training.

Compared to the VR competitors, we're lighter and cheaper: no headset needed, just a browser and a webcam.

Compared to Yoodli/Poised, we're more specific: a subject-aware panel that interrogates actual course content, not generic speech coaching.



Figure 2. Four Actions Framework (ERRC Grid) mapping AlmReady's positioning relative to generic coaches, VR training tools, and the NYU case

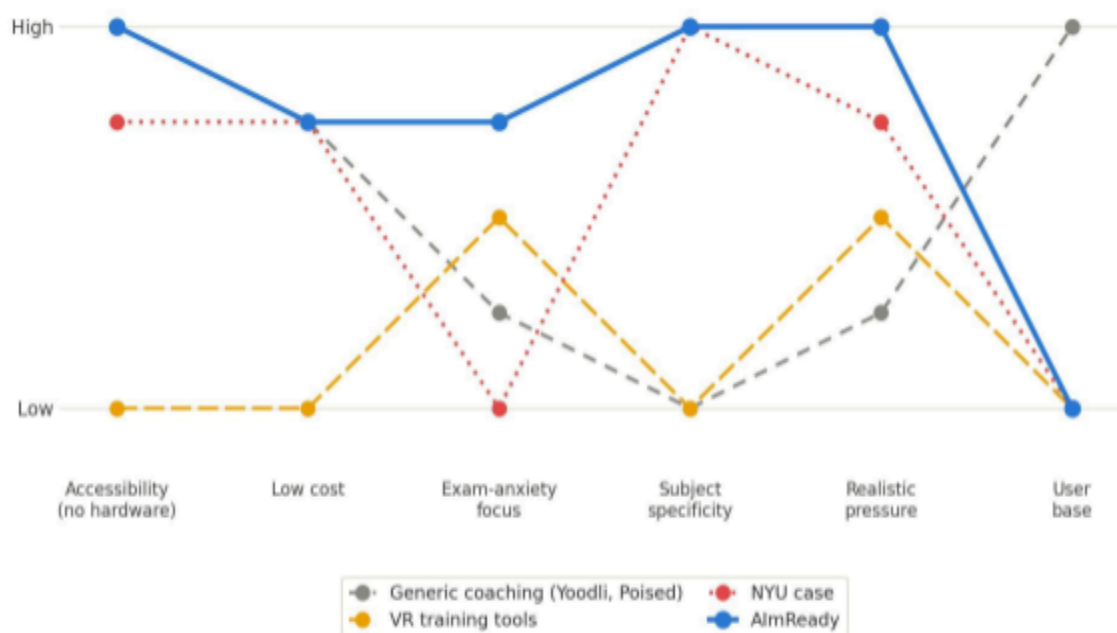


Figure 3. Strategy Canvas - AlmReady matches generic tools on accessibility, exceeds VR tools and the NYU case on exam-anxiety focus and compliance, and is the only solution combining subject-specificity with realistic pressure.

2.3 Validating Competitiveness

We are not entering an empty market: the core mechanism is already public (the NYU case), and larger players already exist (Yoodli, Poised). So the real question isn't "is the idea new?", but "can we realistically compete?". Our answer rests on three points.

First, no existing player occupies our exact position. Direct competitors are generic speech coaches, the NYU case does grading rather than objective training for public speaking, and the VR tools need hardware we don't. As the Strategy Canvas shows (*Figure 3*), we're the only solution that is both specialized for exam anxiety and accessible through a plain browser and webcam.

Second, we're honest that this position isn't permanently safe. Yoodli or Poised could add an "exam mode," since they already have the users and infrastructure. Our defensibility against these direct and generic competitors is layered:

- **Primary Barrier:** a proprietary interaction dataset, built from consented real exam-practice sessions, that compounds over time and cannot be replicated without the same user base.
- **Secondary Barrier:** a product built specifically around oral-exam anxiety and subject-aware interrogation, rather than generic public-speaking coaching.

We must be equally honest about timing: neither barrier exists on day one. The dataset only becomes an advantage as usage accumulates, and specialization alone can be imitated. This means our first year is a window of maximum exposure, in which we are visible to competitors but not yet defended (this is exactly failure cause 2 in Section 6.2). Our plan for that window is speed: grow the free user base aggressively to feed the dataset, and lock in the design-partner university as a reference case before any imitator can. In other words, in Year 1 our defense is not a barrier but a head start. In strategy terms, this is a classic cold start problem: a data-driven network effect only creates value above a critical mass of users. Following the Allee threshold model, if the user base stays below that critical level, the asset never becomes a real barrier. This is also why our free tier exists: a freemium entry is one of the standard tactics to reach critical mass quickly and get past the cold start phase.

Third, our strength is exactly where they're weak. A generic coach can't question a student on the real exam topics, change their follow-ups based on what they just said, or recreate the pressure of a real panel. That kind of subject-aware, challenging practice is the hard part to copy.

In short, we compete not by owning the idea, but by being the most specialized option in a space where everyone else is too generic, too heavy, or aimed at a different goal.

2.4 Industry Forces

We evaluate the sector through Porter's five forces (with the sixth force of Complements):

- **New entrants (high):** the core mechanism (a voice AI examiner plus a multi-LLM panel) is already public through the NYU case [13], so there is no patent barrier. Anyone with API access to a voice model and an LLM can legally build the same architecture; the barrier is not the technology but the proprietary dataset and subject-specific focus.
- **Buyer power (mixed):** on the B2C side, students face near-zero switching costs and have free alternatives, keeping their bargaining power high. On the B2B side, universities are budget-constrained and slow to decide, but once a license is signed and onboarding is complete, switching costs rise, lowering their bargaining power over time.
- **Substitutes (high):** free or low-cost alternatives are numerous and already normalized. Unstructured practice (mirror, family) costs nothing and is the default students already fall back on. Paid alternatives (generic AI speech coaches and VR-based trainers) compete more directly on budget, but none of them target oral-exam anxiety specifically. Either way, the threat is high: students already have a habit of "solving" this problem for free, which is what we have to overcome.
- **Supplier power (high):** our cost structure is heavily concentrated on a single external provider. ElevenLabs' voice generation alone accounts for about 90% of the marginal cost of a paid session (Section 3.2.3), so any price change on their side flows directly into our margins. Model providers (OpenAI, Groq, Anthropic) add a secondary, less concentrated dependency.

- **Rivalry (medium-high):** direct competitors (Yoodli, Poised) are generic and not built around exam interrogation, but they have funding, users, and infrastructure that would let them add an "exam mode" quickly. The NYU case is grading-focused, not training-focused, so it does not directly compete for our anxiety-reduction positioning, but it validates the same underlying mechanism as feasible.
- **Complements (high):** AI infrastructure providers (OpenAI, Groq, Anthropic, ElevenLabs) are also potential future competitors, but today they behave as complements. Building on their models lets us avoid the cost of training foundation models from scratch, in exchange for the supplier-power risk described above.

These forces push our defensibility away from the algorithm itself (which is public and replicable) and toward the proprietary interaction dataset and university relationships, consistent with the positioning argument in Section 2.3.

3. Business Model Canvas

3.1 Core Components Overview

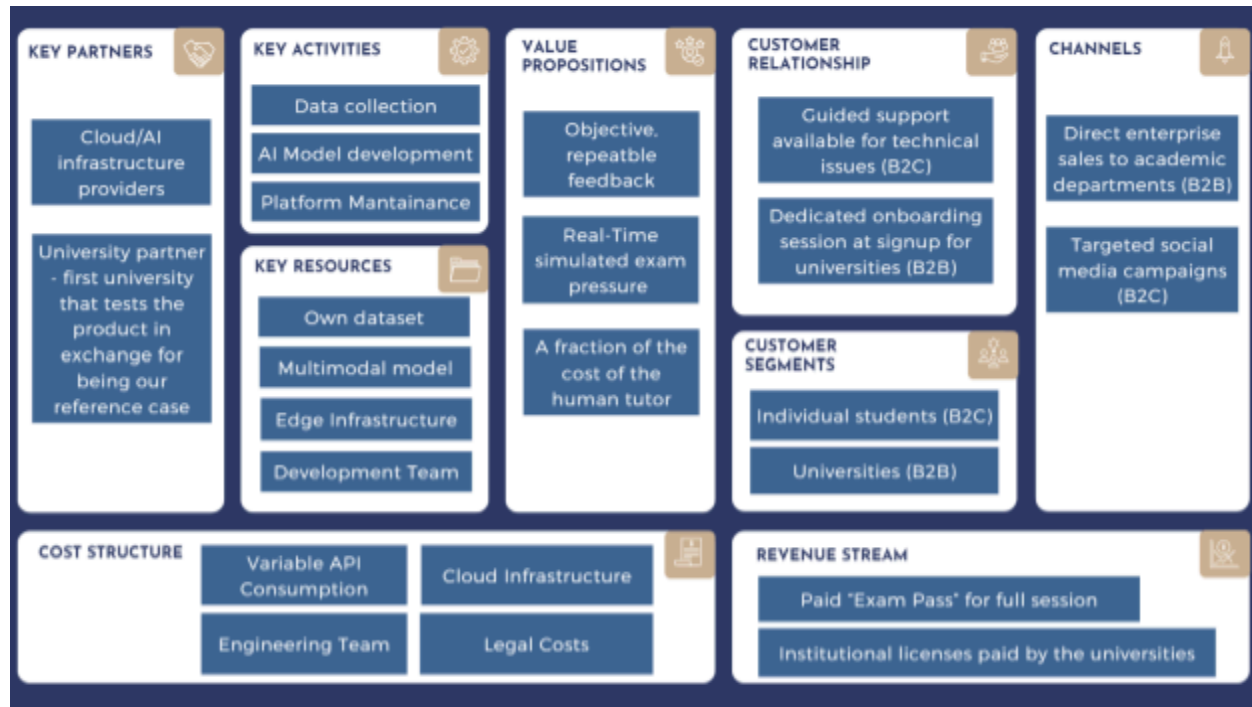


Figure 4. Business Model Canvas

AlmReady serves two **customer segments**: individual students who use the app on their own and universities who buy it as an institution. For students, the **value** is objective and repeatable feedback combined with real-time simulated exam pressure and a cost that's a fraction of what a human tutor would charge. For universities, the value is different: fewer students failing their oral exams and improved overall quality of services offered to students.

We reach these two segments differently (**channels**). For students, we use targeted social media campaigns. For universities we rely on direct enterprise sales to academic departments.

Students get guided support if they run into technical issues and universities get a dedicated onboarding session when they sign the license (plus ongoing assisted support). (**customer relationships**)

Behind the product, we depend on a few **key partners**:

- cloud and AI infrastructure providers, who power the whole system;
- a university design partner (the first university that tests the product in exchange for being our reference case).

Our **key activities** are:

- data collection;
- AI model development, continuously training the multimodal behavioral models and refining the multi-agent examiners;
- platform maintenance, keeping the browser-based pipeline running with low latency and strict data privacy.

Our **key resources** are our own growing dataset, the multimodal model itself, the edge infrastructure and our development team.

On the money side, we have two **revenue streams**:

- a paid "Exam Pass" for full sessions with video and voice;
- institutional licenses, paid by the university per student per year.

Our **costs** are mostly variable API consumption (pay-per-use for speech-to-text, text-to-speech, and multi-agent LLM inference, where text-to-speech alone makes up about 90% of the cost per session) plus cloud infrastructure, engineering team and legal costs (to stay inside the AI Act's regulation for training tools).

3.2 AI-related Assets Focus

Focusing on our AI assets (data, models and infrastructure) is essential to ensure both a robust technical system and a profitable business model.

3.2.1 Data Assets and Proprietary Moat

Initial model training is bootstrapped utilizing validated public datasets:

- TED-LIUM [20] for speech recognition,
- a recent 2026 multimodal dataset of student oral presentations [21].

Over time, we build our own dataset from real session data, always with the user's explicit consent. The dataset grows on two levels: free sessions contribute conversational data (transcripts, pauses, filler words), while paid sessions add the richer behavioral metrics extracted from the webcam (gaze direction, gesture stability). In this way every consented session, free or paid, improves our models.

To guarantee privacy and regulatory compliance, all raw video footage is permanently deleted. The system only saves and learns from the extracted numerical scores.

3.2.2 Edge Computing and Privacy Architecture

The visual processing pipeline executes entirely on the user's edge device via browser-based computer-vision frameworks (e.g., MediaPipe [22] or TensorFlow [23]). The system is engineered to transmit only lightweight scalar values to the backend (such as gaze-deviation

scores or head-movement velocity metrics), never raw video feeds or facial imagery. This privacy-by-design approach has already been studied in research on systems that monitor drivers. That study looked at exactly this same trade-off: deciding which parts of facial-expression processing can stay on the device, and which parts, if any, need to be sent to the cloud [24]. We apply the same logic here: all facial-expression processing stays on the student's device, and only small numeric scores reach our servers.

By never transmitting raw biometric data, the system ensures full compliance with the European AI Act (Section 6.2).

3.2.3 Infrastructure Cost and Tier Differentiation

To base our pricing strategy on real numbers, we calculated the cost of a 30-minute exam session (20 minutes of student speech and 10 minutes of AI speech). Using July 2026 API prices from OpenAI [25], Groq [26], and ElevenLabs [27], the cost breakdown is:

Component	Technology & Details	Estimated Cost
Video Analysis	Processing happens entirely on the user's device	\$0.00
Speech-to-Text	Whisper (20 minutes of student speech)	~\$0.037
AI Conversation Logic	- GPT-4o-mini (lead examiner) - Llama 3.1 8B (assistant) - Claude Haiku (silent coordinator)	~\$0.013
AI Voice Generation	ElevenLabs (10 minutes of AI speech)	~\$0.50
Grading council (runs after the session)	GPT-4o-mini + Llama 3.1 8B, rereading the transcript	~\$0.005
Total Cost per Paid Session		~\$0.55

This cost structure directly shapes our free and paid plans:

- The **Free** Tier: Focuses on text and basic audio. It uses a single AI agent and the browser's free, built-in voice. This keeps the cost around \$0.04 ($=\$0.037+0.004$ for a single agent), allowing us to offer a generous number of free practice sessions (capped at 20 per year, see Section 3.4) at almost no cost.
- The **Paid** "Exam Pass" Tier: Unlocks the webcam behavioral analysis, the full three-agent panel, the realistic ElevenLabs voice and the after-session report. The actual cost of \$0.55 per session is comfortably covered by the user's subscription fee.

3.3 Assumption and interdependencies

The viability of AlmReady relies on a few critical assumptions and technical dependencies.

First, we assume that students are willing to pay for an app that aims to reduce exam anxiety and that universities will buy the B2B solution to improve student retention and performance. We also assume that our multi-agent simulation is realistic enough to recreate actual exam pressure, as a simpler chatbot experience would not trigger the psychological exposure needed to reduce anxiety.

Technically, our solution relies on two critical factors. First, we must minimize API token costs to ensure that our freemium business model remains profitable. Second, we need real-time, low-latency synchronization between voice transcription, computer vision, and the AI agents. Any delay in processing these data streams would break the natural flow of the conversation and ruin the realism of the exam simulation.

3.4 Feasibility and Scalability logic

The platform's scalability relies on two structural choices: an edge architecture and a cost-driven freemium split.

Because all video processing runs client-side, the paid tier's webcam analysis, the most computationally expensive task, never touches our servers, removing the need for GPU-heavy cloud infrastructure. In this way the backend is only responsible for lightweight text routing and orchestration of the Speech-to-Text, Large Language Model, and Text-to-Speech calls. This allows the system to support thousands of concurrent sessions with minimal infrastructure cost and it means that offering video analysis as a paid feature is a product decision rather than a cost-driven necessity.

The freemium split then isolates the cost components that do not scale for free: mainly the high-fidelity voice generation, which accounts for over 90% of a paid session's marginal cost (~\$0.50 of ~\$0.55).

Paying users also generate a small additional cost after the session ends: the same GPT-4o-mini and Llama 3.1 8B models reread the full transcript to produce the final feedback report, adding about \$0.005 per session. Because this asynchronous review runs after the live exam, it does not affect latency or the real-time experience, and it keeps the total paid-tier cost close to \$0.55 per session.

By keeping this full premium experience strictly as paid-access content, the free tier can stay capped at a low number of sessions (e.g. 20/year) at almost no cost, giving students an easy way to start practicing with text and audio only. The paid tier has a small and predictable cost per session, which is easily covered by the subscription price, since it is directly associated with the platform's most advanced capabilities (realistic exam simulations and comprehensive behavioral feedback).

Each subscription pays for the specific costs it generates. As the number of paying users grows our variable costs grow with them but always in the same proportion, since every paid session generates more revenue than it costs. This direct link between cost and revenue is what makes the model sustainable at scale.

3.5 Suggested Pricing

Tier	Price	Why
Free (With a limit on the maximum number of session per year)	€0	Real cost ~\$0.04/session.
Exam Pass (5 full sessions)	€8-9	Real cost ~\$2.75 (5 × \$0.55), ~65-70% margin.
Institutional license	~ €3/student/year	This institutional model allows students to access the platform at no cost, while guaranteeing us strong profit margins (see Section 4.2.3).

(per-session costs assuming a ~1:1 USD/EUR rate for simplicity)

4. Addressable Market Analysis

4.1 Market Size Components

The market is divided into three levels:

- The Total Addressable Market (TAM) includes all tertiary students in Europe.
- The Serviceable Available Market (SAM) includes all Italian students enrolled in fields where oral exams are prevalent.
- The Serviceable Obtainable Market (SOM) includes the first-year Italian students in these target fields that can realistically be reached and converted during the first year of launch.

Since our business model has two distinct revenue streams, we size them separately. In the B2C stream, the revenue unit is the individual paying student, because students purchase the premium access based on their specific exam session needs. In the B2B stream, the revenue unit is the institutional license, paid by the university per enrolled student.

The B2C formulas are therefore:

$$\text{TAM} = (\text{Tertiary Students in Europe}) \times (\text{Revenue per Student})$$

$$\text{SAM} = (\text{Target Students in Italy}) \times (\text{Revenue per Student})$$

$$\text{SOM} = (\text{Converted First-Year Students in Year 1}) \times (\text{Revenue per Student})$$

The B2B stream is sized with the same three-level logic, using the institutional price per enrolled student as the revenue unit.

4.2 Estimation Methodology

A top-down approach starts from a broad industry figure and applies a market-share percentage to it, while a bottom-up approach starts from the individual units of the business (pricing and expected customer behaviour) and aggregates them into a total [28]. We chose the bottom-up method because it builds market size from real customer data rather than a single assumed percentage [29], which validates the estimate in concrete numbers instead of a guess. Investors also tend to trust this approach more, since each assumption can be tested individually.

4.2.1 Students in Europe, Italy, and Reachable Users

TAM: Our Total Addressable Market includes all university students in Europe, since any student can technically use our browser-based tool. According to official European data [30], there are 19.1 million university students in Europe.

SAM: To find our Serviceable Available Market, we focus on Italy. There are approximately 2.3 million university students currently enrolled in Italy [30].

However, since our AI is explicitly designed to simulate oral exams, our primary market excludes STEM disciplines, which rely heavily on written tests. Instead, we focus on fields like Law,

Medicine, Humanities, and Social Sciences where oral examinations are the standard assessment method. Eurostat data shows that these specific fields make up 62% of Italian university students [31]. By applying this percentage, our target market in Italy is about 1,400,000 students.

SOM: For our first year of operations, we will not target all the students. Instead, we will only focus on first-year students in those specific fields. We chose this group because new students experience the highest anxiety about oral exams and actively look for new study tools to define their learning method. In marketing terms, this is a segmentation and targeting choice: we divide the market by field of study and year of enrollment and then apply a focused (niche) targeting strategy, concentrating our limited resources on the single segment where the need is strongest and the users are easiest to reach.

Italy registers about 330,000 new students every year [32]. Applying our 62% filter to this group gives us a realistic target of about 200,000 first-year students.

1 Year estimation: Before applying conversion rates, we subtract the students already covered by the institutional license of our design-partner university (University of Pisa, see Section 4.2.3): its 9,000 first-year students [33], filtered by the same 62%, give 5,580 students who will access the tool through the institutional license and therefore will not pay in the B2C channel.

The effective B2C base becomes: $200,000 - 5,580 = 194,420$ students.

To estimate how many paying users we will actually acquire from this effective base, we combined marketing projections with established SaaS industry benchmarks [34]. We project that our targeted, seasonal campaigns (concentrated immediately prior to exam sessions) will reach 30% of these 194,420 students. Out of the students reached, we estimate 25% will sign up for a free account, a target that aligns well with the standard conversion rate for landing pages. Finally, 4% of these free users will upgrade and buy the paid "Exam Pass," which perfectly reflects the standard free-to-paid conversion range typical of freemium business models.

Converted Students = $194,420 \times 30\% \times 25\% \approx 14,500 \times 4\% \approx 583$
(rounded to ~600 paying users).

The remaining ~14,000 signed-up students are not a wasted audience: they build the conversational layer of the proprietary dataset (transcripts, pauses, filler words), while paying users add the multimodal behavioral layer (Section 3.2.1).

4.2.2 Revenue per Student

Revenue per student is estimated separately for the two channels, since they have different pricing logs.

B2C revenue per paying student (ARPU: Average Revenue Per User) is based on the real pricing of the "Exam Pass" (€8 for 5 full sessions, Section 3.5), rather than a generic monthly subscription. Since students study around distinct academic periods, purchases are tied to the two main Italian university exam sessions (winter and summer):

- A light user buys 1 pass (€8/year)
- An average user buys 2 passes (€16/year)
- A heavy user buys 3 or more (€24+/year)

As a working estimate we use the average profile: 2 passes per year, giving us a B2C ARPU of €16/year.

B2B revenue per student is the institutional license price of €3 per enrolled student per year (Section 3.5), applied to all students enrolled in the target-field departments covered by the license, regardless of how many of them actually use the tool.

4.2.3 Final Computation

B2C computation

The three B2C market levels are:

Level	Population	Yearly revenue (at €16 ARPU)
TAM (EU, all fields)	19,100,000 students	~€306M
SAM (Italy, target fields)	~1,400,000 students	~€22.4M
SOM (First Year, target fields)	~600 paying students	~€9,600

To check whether the Year-1 B2C channel sustains itself, we compare its revenue with its direct infrastructure costs (per-session costs from Section 3.2.3, assuming a ~1:1 USD/EUR rate for simplicity). Paying users are assumed to run 10 full sessions per year (2 Exam Passes); free users, having a free and lightly limited tier, are prudentially assumed to run 15 free sessions per year:

- B2C revenue: 600 paying users × 2 passes × €8 = €9,600
- Costs, paying users: 600 × 10 sessions × €0.55 = €3,300
- Costs, free users: ~14,000 × 15 sessions × €0.04 = €8,400
- B2C net margin (Year 1): €9,600 – (€3,300 + €8,400) ≈ – €2,100

The B2C channel operates at a managed loss in Year 1. However, rather than a financial weakness, this is a deliberate strategy. We are effectively spending €2,100 to acquire ~14,000 active users. This negligible cost acts as our data acquisition investment, allowing us to build the proprietary dataset required to train our AI and expand our competitive advantage.

B2B computation

We size the institutional channel by conducting a bottom-up analysis based on our design-partner university. The University of Pisa has about 46,000 enrolled students [33]. The university would pay the license only for the departments where oral exams are prevalent, the same 62% of students used above: 46,000 × 62% = 28,520 covered students.

On the cost side, not every covered student will actually use the tool. Based on published evidence on student adoption of university-provided study services close to exams, we assume

that 28% of covered students actively use the service [35], each running 10 sessions per year (2 Exam Passes):

- B2B revenue: $28,520 \text{ covered students} \times \text{€}3 = \text{€}85,560$
- Active users: $28,520 \times 28\% \approx 7,986$
- Costs: $7,986 \times 10 \text{ sessions} \times \text{€}0.55 \approx \text{€}43,900$
- B2B net margin: $\text{€}85,560 - \text{€}43,900 \approx \text{€}41,700$

Applying the same three-level logic to the institutional channel:

Level (B2B)	Covered population	Yearly revenue (at €3/student)
TAM (EU, all tertiary students)	19,100,000 students	~€57.3M
SAM (Italy, target fields)	~1,400,000 students	~€4.2M
SOM (Year 1: one design-partner university)	28,520 students (Pisa)	~€85,560

Combined Year-1 picture

Stream	Revenue	Direct costs	Margin
B2C – paying users	€9,600	€3,300	+ €6,300
B2C – free tier	€0	€8,400	– €8,400
B2B – one institutional license (Pisa)	€85,560	~€43,900	+ €41,700
Total Year 1	€95,160	~€55,600	+ €39,500

4.3 Investment Decision Support

4.3.1 Cost-Volume-Profit (CVP) and Break-Even Analysis

To check when AlmReady actually covers all its costs, we run the break-even analysis in two steps. First, we compute a per-session Quantity Break-Even Point (QBEP) for the B2C channel alone, since both its revenue (the Exam Pass) and its marginal cost (Section 3.2.3) are defined per session. Then, since AlmReady actually sells through two channels with different margins, we compute the overall break-even for the whole company using the multi-product method, based on the contribution margin ratio weighted by the sales mix.

Fixed Costs (FC): €180,000/year. Bottom-up estimation:

- Core ML/engineering team (multi-agent development, model fine-tuning): €120,000
- Backend orchestration (lightweight, no dedicated GPUs needed thanks to edge-based video processing): €15,000
- External consultant for legal review and AI Act compliance monitoring (Section 6.2): €5,000
- B2C marketing and B2B sales baseline: €30,000

- General & Administrative: (software licenses, accounting, and basic operational overhead): €10,000

Step 1: B2C break-even per session. The Exam Pass sells at €8 for 5 sessions, giving a price per session $p = €1.60$. The variable cost per session is $v = €0.55$ (Section 3.2.3). The B2C Unit Contribution Margin ($CM_u = p - v$) is €1.05, a Contribution Margin Ratio of about 66%, consistent with the 65–70% margin stated in Section 3.5. The break-even volume is:

$$QBEP = FC / (p - v) = 180,000 / 1.05 \approx 171,400 \text{ sessions/year}$$

This number is a useful reference, but it is not realistic on its own: it assumes the whole company is sustained by B2C sessions, while Chapter 4.2.3 shows that most Year-1 revenue comes from the institutional channel.

Step 2: Contribution by channel. At the Year-1 volume projected in Section 4.2.3, paying B2C users (600 users $\times$ 10 sessions $\approx$ 6,000 sessions) generate a contribution of about €6,300. The free tier, however, costs about €8,400/year to run, so the B2C channel as a whole has a negative net contribution of about –€2,100, consistent with the managed loss already discussed. The Pisa license covers $\approx 79,860$ sessions/year (28,520 covered students $\times$ 28% active usage $\times$ 10 sessions each), costing about €43,900/year against a fixed revenue of €85,560/year: a B2B contribution of about €41,700/year. Combining both channels, Year-1 total net contribution is about €39,500/year.

Step 3: Multi-product break-even. Since the two channels have different margins (66% B2C, 49% B2B), we compute a single overall CM ratio by putting together their totals: the two channels generate a combined contribution of €48,000 on a combined revenue of €95,160, giving an overall CM ratio of about 50%. In other words, in the Year-1 sales mix, every euro of revenue leaves about 50 cents to cover fixed costs. The free tier costs €8,400/year but generates no revenue, so it cannot enter the formula as a variable cost: we treat it as a fixed cost of the strategy (frozen at the Year-1 level), bringing total fixed costs to €188,400. This is a deliberate simplification: it holds only under the assumption that the free user base stays close to the Year-1 estimate. If free sign-ups grow substantially, this "fixed" cost grows with them, raising the break-even revenue accordingly. The company-level break-even is therefore:

$$\text{Break-even revenue} = 188,400 / 0.50 \approx €374,000/\text{year}$$

Keeping the same sales mix, this corresponds to about four Pisa-sized institutional licenses (at the Year-1 mix, B2B accounts for $\sim 90\%$ of revenue: $€374,000 \times 90\% \approx €336,000$, and $€336,000 / €85,560 \approx 4$ licenses). In other words, the startup needs to acquire about 3 more licenses beyond Pisa (Section 6.2, risk 4). Winning these licenses also raises AlmReady's visibility, which is likely to drive more B2C sign-ups and paid conversions than the Year-1 estimate; if B2C growth is strong enough, it could cover the free tier's cost and add its own margin on top of the institutional channel.

4.3.2 Strategic Implications of the Market Size

These estimates are not intended as exact predictions: they are transparent calculations based on explicit assumptions. Four conclusions follow directly from the numbers:

1. **The market is large**, even restricting to the Italian target fields, the serviceable market is worth roughly €22M/year on the B2C side and €4M/year on the institutional side. The European TAM is significantly larger, representing a much broader market opportunity.
2. **Operating at a loss in Year 1 is normal, not a warning sign.** Academic research on startup finances shows that profitability doesn't grow in a straight line. In the early stage, costs usually grow faster than revenue, and profitability only stabilizes later, once the business matures [36]. AlmReady's negative net contribution in Year 1 (Section 4.3.1) follows this same pattern. In Year 1, the goal is not to make profit. The goal is to build our free user base and collect proprietary interaction data, which is our main defensible asset (Section 2.3).
3. **A realistic path to break-even is reachable within this typical window.** The break-even analysis (Section 4.3.1) shows that closing the €140,400/year gap requires roughly 3 to 4 more institutional licenses beyond Pisa. We plan to pursue this in Year 2, targeting 3 additional university licenses on top of Pisa, while continuing to grow the B2C base. Each institutional deal also increases AlmReady's visibility, which should help B2C base to grow fast.
4. Assuming the demand is there (the validation strategy is meant to confirm it), **the main bottleneck is conversion speed, not the size of the market.** On the B2C side, students care about exam preparation only during narrow, highly seasonal pre-exam windows, so reaching and converting them efficiently in those windows is the key operational challenge. On the B2B side, universities are slow, budget-constrained buyers, so the length of the institutional sales cycle is the main risk to the Year-1 margin. Both bottlenecks, and the conversion rates assumed above (30% reach, 25% sign-up, 4% paid conversion, 28% institutional usage), are exactly what our validation strategy is designed to test with real data before committing further resources.

Overall, the analysis confirms that the opportunity justifies the investment: it is normal for a venture at this stage to operate at a loss in Year 1, and even a single institutional license already makes the opportunity worth pursuing. Reaching full break-even is a realistic goal for Year2, provided we secure about 3 additional institutional licenses and the B2C base continues to grow.

5. Personas and Validation Strategy

Chapter 4 showed that our market estimates rest on a small set of explicit assumptions: the reach of our campaigns (30%), the sign-up rate (25%), the free-to-paid conversion (4%), and the institutional usage rate (28%). This chapter defines who we must learn from (the personas) and how we will test those assumptions with real people, cheaply and quickly, before committing significant resources.

5.1 User Personas Definition

Our product reaches different people who use and buy it for different reasons, so we define three personas: two on the B2C side, the individual students who pay out of pocket, and one on the B2B side, the university that signs an institutional license.

Persona	Role and context	Frustrations	Goals
The Anxious Student (B2C buyer and user)	First-year university student in a target field (Law, Medicine, Humanities, Social Sciences), preparing for an oral exam, a thesis defense or a first job interview. Usually practices alone, in front of a mirror or with family members. Limited budget.	Practicing with a mirror or family gives no real feedback; human tutors are expensive; existing apps feel like passive Q&A and do not recreate the pressure of a real interrogation.	Find a safe space to fail without consequences; receive concrete, objective feedback on content and delivery; learn to manage pre-exam panic through repeated, realistic practice.
The First-Timer (B2C buyer and user)	First-year student in a target field about to face the first oral exams of their life. Coming from high school, where tests were short and frequent, they have never prepared a full oral exam and have no study method for it yet. Not anxious, but unsure whether they actually know the material.	Doesn't know what a university oral exam looks like: what questions to expect, how deep the answers must go, how long to speak. Studying alone gives no way to check whether the material is truly mastered or just familiar, and friends or family can't judge subject accuracy.	Understand what examiners expect before sitting the real exam; test their preparation in advance and find weak topics while there is still time to fix them; experience the format of an oral exam at least once, instead of learning it by failing.

The University Decision-Maker (B2B buyer, not a user. Since in an institutional purchase the buyer and the user are different people, our B2B validation must target the buyer, because they are the ones who decide whether the license is signed.)	Institutional figure responsible for teaching quality and student services (e.g. department director). Reviews the same statistics every session: fail rates, students falling behind, average time-to-degree. Manages a limited budget for student support services.	Sees the problem clearly in the data (failed exams, delays, dropout) but has no scalable tool to help anxious students before exams; individual tutoring does not scale; any tool that touches student data raises immediate AI Act concerns.	Improve measurable institutional outcomes (pass rates, retention, time-to-degree) with a scalable service, without changing how professors teach or grade, and without taking on regulatory risk.
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5.2 Validation Strategy Methods

We validate step by step, ordered from cheap and fast to slow and expensive, so that each step risks only the resources justified by the evidence of the previous one. Every method targets a specific assumption from previous chapters.

Method	What we do	Assumption tested
1. Problem interviews	Short interviews with ~30 students in target fields and ~5 university staff involved in student services.	The problem is real and felt: exam anxiety and lack of realistic practice.
2. Surveys	Online survey to ~150 students and a shorter one to university staff, testing interest in the specific solution and willingness to pay (Exam Pass price points).	Demand intensity; the 28% usage assumption; price acceptance of €8 per pass.
3. Landing page pre-launch	A single-purpose landing page describing the product, with a waitlist sign-up, promoted to students via paid social campaigns during exam season. Target: 500 qualified visits in 4 weeks.	The 25% sign-up assumption and, indirectly, the feasibility of the 30% reach plan.
4. MVP	A working prototype limited to the core loop: text/audio oral simulation with one examiner persona and a basic feedback report, released to waitlist users. This also facilitates the improvement of the dataset with new real interaction data and the training of our AI models.	Students actually use it repeatedly and report increased confidence (pre/post surveys), validating the value proposition.

5. Pilot with the design-partner university	Run the full MVP for 8-12 weeks inside our design-partner university (University of Pisa), under a formal agreement with the institution.	Students who use the tool show better exam outcomes than a comparable group who don't. The pilot also gives us a first real measurement of the usage rate.
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5.3 Learning and Decision-Making

Each validation step has explicit success and failure signals defined before the test starts, so that decisions are driven by evidence rather than by attachment to the idea. The thresholds obtained by specific benchmarks are:

Test	Failure signal (stop / rethink)	Positive signal (proceed)	Strong signal (accelerate)
1. Problem interviews	Below 30% confirm the problem as significant	30-60% confirm it	Above 60% confirm it and describe current workarounds
2. Surveys	Below 25% interested in the solution	25-50% interested	Above 50% interested, with willingness to pay at €8
3. Landing page	Below 15% of visitors leave contact information	15-25% leave contact information	Above 25%, a strong signal toward the 25% target
4. MVP usage	Users try once and never return; no confidence improvement in pre/post surveys	Repeated use; measurable confidence increase	Students recommend the app to each other; users request the full paid features
5. University pilot	Low adoption and usage rates; the university does not renew interest.	Strong adoption and usage rates in pilot groups; the university continues the collaboration.	High, sustained usage and conversion of the pilot into the first paid institutional license. Searching for a new university.

Three decision rules govern how we act on these signals.

Persevere. If a step returns a positive or strong signal, we proceed to the next, more expensive step and the measured value replaces the corresponding assumption in the Chapter 4 model. For example, the real landing-page sign-up rate replaces the assumed 25% and the real pilot usage rate replaces the assumed usage of 28%: the financial model is progressively re-estimated on primary data instead of benchmarks.

Pivot. If the problem is confirmed but the solution signal is weak (e.g. interviews strong, MVP usage weak), we change a specific element rather than abandoning the idea: pricing, feedback format or target departments. When we pivot, we don't restart everything, we go back only to the cheapest test that can check the specific change we made.

Stop. If fewer than 30% of interviewed students confirm the problem, the demand hypothesis itself is wrong and no product iteration fixes it. If the pilot shows no usage rate improvement or students aren't using the tool (so that the quality of service the university offers its students doesn't improve), the B2B value proposition collapses. Only the B2C channel would survive, and Chapter 4 shows it is not economically sufficient on its own. Defining these kill conditions in advance is what makes the validation honest: we specify today what evidence would prove us wrong.

The overall logic is that every investment is released only after the previous cheaper test has reduced the corresponding risk. By the end of the ladder, the assumptions of Chapter 4 are no longer estimates taken from benchmarks: they are either measured values or falsified hypotheses, and in both cases the decision rests on evidence.

6. Failure Analysis and Risk Mitigation

6.1 Negative Backcasting Method

In this chapter we deliberately reverse our perspective. Instead of asking "what could go wrong?", we assume that AlmReady has already failed two years from now and we work backward to reconstruct the most plausible reasons why. This technique, known as negative backcasting, helps to overcome natural optimism. By treating failure as an event that has already happened, rather than a remote risk, we can clearly identify and discuss weaknesses that we might otherwise ignore.

The method has three steps. First, we imagine the concrete failure scenario. Second, we list the most credible causes that could have produced it. Third, for each cause, we define how to spot the problem early and how to fix it. This turns a simple list of fears into an actionable monitoring plan.

6.2 Common Failure Causes

Working backward from the imagined failure, we identified six credible causes, ordered by severity.

1. Regulatory prohibition

While Article 5(1)(f) of the EU AI Act strictly bans emotion-inferring AI in education [37], AlmReady operates outside this ban. We do not attempt to recognize or profile emotional states; our system detects objective physical gestures and behavioral cues commonly associated with exam stress.

The European Commission's guidelines clarify that role-play and training simulations, where the AI detects behavioral signals without the result affecting a student's evaluation or certification, fall outside the ban [37]. AlmReady's legality in the EU therefore depends entirely on staying on the "training, non-certifying" side of that line.

However the failure scenario is concrete: AlmReady operates at the absolute borderline of the current EU regulatory framework. As the legal landscape is still evolving, future shifts in the Commission's interpretations or stricter amendments by national authorities could suddenly narrow the 'training exception'. If the regulatory boundary moves, our core behavioral analysis features could suddenly be reclassified as non-compliant. This risk is existential because a sudden reclassification would force us to either completely modify our core AI features or abandon the EU market.

2. Increase of competition

The core mechanism (a voice AI examiner plus a multi-LLM panel) is already public and unpatentable due to the NYU prior art [13], which means we have no legal barrier to imitation: anyone can legally build the same architecture.

What makes this concrete rather than theoretical is that credible imitators already exist. Established players like Yoodli[14] or Poised[15] already have what takes years to build (users, brand recognition, funding and running infrastructure for real-time speech analysis) and adding an "exam mode" to their existing products would be an incremental feature for them.

Our only durable defense is the proprietary interaction dataset built from consented sessions, but this asset has a critical property: it grows with usage and therefore with time. In the first year, when the dataset is still small, the asset barely exists, so there is a window in which we are most visible and least defended, because the dataset is not yet an advantage (the cold start problem described in Section 2.3). This is especially true for the multimodal behavioral layer, which only grows with paid sessions and is therefore the slowest part of the dataset to accumulate.

The failure scenario is a well-funded replication of our product, using their existing user base to reach scale faster than we can, and closing the market before our dataset becomes a real barrier.

3. Market adoption failure (B2C)

The Chapter 4 is based on conversion assumptions (30% reach, 25% sign-up, 4% paid conversion) that could simply prove wrong. Each of the three rates carries a different kind of fragility:

- The 30% reach is a planning assumption, not a benchmark: it depends on our marketing budget and on how efficiently paid social campaigns can target students in our specific fields.
- The 25% sign-up rate is anchored to landing-page benchmarks, a weak page or cold audience could cut it in half.
- The 4% paid conversion is a standard freemium number, but it's borrowed from products people use all year round. Ours is seasonal: students mainly use it in the weeks before exams, so we can't be sure the same conversion rate holds with a different usage pattern.

Because the three rates multiply, errors compound: if each one turns out just moderately lower than assumed (say 20%, 15%, and 3%), the ~600 paying users of Year 1 collapse to fewer than 180, cutting B2C revenue by more than two thirds.

The failure scenario is a launch that lands between exam sessions, converts below thresholds at every step and exhausts the marketing budget before the next demand peak arrives.

4. Institutional sales failure (B2B)

Chapter 4 showed that the institutional channel is the profit engine: the B2C channel alone operates at a managed loss, so the project's sustainability rests on institutional licenses. Our model assumes that, before launch, the design-partner university has signed a conditional agreement: a formal commitment to purchase the license if the pilot meets the success thresholds defined in Section 5.3. We do not start without this signed commitment, but the payment itself is triggered only by the pilot's results.

The failure here is an inability to grow beyond that single university. Possible factors are:

- The first deal may be unrepresentative: our design-partner university has a direct relationship with us and a reason to support the project, advantages that no second university shares.
- University buying is structurally slow: budgets are decided on annual cycles, approvals pass through multiple layers, and the people who see the results are usually not the people who sign the license, so demonstrating value can be difficult.

The failure scenario here is a pre-committed license that produces a strong reference case, but then stops: the second and third universities move too slowly or never convert.

5. Technological failure

Two technical risks could deteriorate the product's core value, one about speed and one about quality.

Latency. The realism of the exam simulation depends on real-time synchronization between transcription, vision analysis and the multi-agent panel. Each step in this chain adds delay, if the total delay grows beyond the rhythm of natural conversation, the examiner appears to "think" for seconds before every question, the interruptions and follow-ups questions that create pressure become impossible and the simulation stops feeling like an interrogation and starts feeling like a chatbot, eliminating the psychological pressure that is our core value proposition.

Performance. If the models do not convincingly simulate a real exam commission, students cannot receive objective and reliable feedback. This can fail in two ways:

- The examining agents may ask questions that are too generic, off-topic, or disconnected from what the student just said, breaking the illusion of being questioned by someone who is actually listening.
- The evaluation system may produce inaccurate or inconsistent feedback, judging the same answer differently across sessions.

Both failures destroy trust quickly: a student who receives one obviously wrong question or one unfair report has little reason to come back, and in a product whose value is built on repeated practice, losing users after the first session means losing them entirely.

6. Cost-structure

Voice generation alone accounts for about 90% of the marginal cost of a paid session (~€0.50 of ~€0.55, Section 3.2.3). This concentration means our unit economics depend heavily on the pricing decisions of a single external provider: a price increase on the voice API would flow directly into our cost per session.

The same margin is exposed from a second direction: usage. Our cost model assumes paying users run 10 sessions per year and free users about 15. Because sessions are not individually billed to the user (the pass covers 5 full sessions, the free tier is limited but open), users who practice far more intensively than estimated raise our costs without raising our revenue. The

free tier is the most sensitive point: it exists to feed conversions and the dataset at a cost of about €0.04 per session, but that cost is only negligible as long as average volume stays near expectations. If the number of users using the free tier grows several times beyond the estimate, it becomes a real success, but too expensive to afford.

The B2B channel is exposed to the same kind of risk. The institutional license is a fixed fee per student, so our revenue from a university is locked in regardless of usage. Our cost, however, is not: we assumed that 28% of covered students actively use the service. If real usage runs higher, the number of sessions we have to serve rises while the revenue stays exactly the same.

The failure scenario is a slow margin decrease: costs rise per session and per user while prices stay fixed, until the model that looked sustainable on paper no longer covers its own variable costs.

6.3 Early Warning Signals and Mitigation

For each of the six causes above, we define in advance a signal we can watch for, and the action we would take if it appears. Defining these before launch is what turns the failure analysis into an actual monitoring plan, rather than a list of fears.

Failure cause	Early warning signal	Mitigation
1. Regulatory prohibition	A Commission updates a national amendment	Get an official legal review from an AI Act specialist before selling to any EU university; keep a fallback mode ready: if the exception narrows, we drop the video and body-language analysis (the contested part) but keep the audio-based features that don't infer emotions, like counting hesitations, repeated words, and long pauses to help students improve their oral delivery
2. Increase of competition	An established competitor (e.g. Yoodli, Poised) announces an "exam mode"	Grow the proprietary dataset aggressively from day one; deepen the one thing they don't have: a subject-aware panel focused specifically on oral-exam anxiety, not generic speech coaching
3. Market adoption failure (B2C)	Landing-page sign-up or paid conversion tracking well below the 25% / 4% thresholds during the first campaign	Concentrate all marketing costs on pre-exam windows when willingness to pay is increased; keep the free tier limited enough to protect conversion; re-estimate the model on real rates before scaling the marketing budget

4. Institutional sales failure (B2B)	The second and third universities stall in procurement, or the design-partner success story doesn't attract interest from other universities	Use the pre-committed first license to produce a hard reference metric (pass-rate / anxiety data); start second-university conversations during the first pilot, not after
5. Technological failure	Users drop out mid-session; feedback that the examiner "lags" or asks off-topic questions; inconsistent reports across sessions; negative feedback from optional user reviews	Fall back automatically to audio-only when the connection is weak; lock question relevance to the uploaded material; analyze carefully user reviews to understand where to invest
6. Cost-structure	Voice-API price increase; free-tier/university usage growing several times beyond the estimations	Keep a cheaper Text-To-Speech provider ready as a fallback, accepting a temporary loss in voice realism while renegotiating or migrating; limit free-tier sessions per user

6.4 Strategic Thinking Benefits

Deliberately reasoning from failure changes decisions we make today, not just the ones we make if things go wrong. Three examples from this chapter show how.

First, naming regulatory reclassification as our top existential risk is precisely why the product is framed from the start around objective behavioral signals rather than emotions and why an audio-based fallback mode is part of the design from the beginning: if the exception ever narrows, we can drop the video/body-language analysis and keep the audio features that don't infer emotions (counting hesitations, repeated words, and long pauses) to help students improve their oral delivery. We build the escape route before we need it.

Second, admitting that the core mechanism is unpatentable (risk 2) is what pushes our entire strategy toward the real advantages we can actually defend (our proprietary dataset and our narrow focus on exam anxiety) instead of relying on an idea anyone can legally copy.

Third, recognizing that our margin depends almost entirely on one external voice provider (risk 6) is why we treat provider substitutability and free-tier limits as design decisions, not accounting details.

This kind of analysis matters even more for an AI startup, where the technology is strong but the legal and competitive situation keeps changing. Deciding in advance what evidence would show us the project is failing is what keeps the plan realistic: it forces us to spot bad news early, while there's still time to react.

7. References

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